

Poisson Regression

Estimation, Interpretations and Diagnostics

Last week: Logistic Regression

- Logistic regression: definition
- Estimation and Interpretation for different type of covariates
- Inference: confidence intervals and tests
- Fitted values and residuals
- Diagnostics: overdispersion

This week: Poisson Regression

Poisson regression is a regression analysis designed for modelling **count data**.

The response variable represents the number of times an event occurs in a given timeframe or even in a space such as a geographic unit. For example,

- the number of customer arrivals
- disease occurrences
- traffic accidents

The response

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Poisson is a common distribution to model a random variable that takes discrete non-negative integer values (i.e., 0, 1, 2,...)

$$Y_i | \mathbf{X}_i \sim \text{Poisson}(\lambda_i),$$

where λ_i is the conditional mean.

A particularity of the Poisson distribution is that its mean equals its variance:

$$\lambda_i = E[Y_i | \mathbf{X}_i] = \text{Var}[Y_i | \mathbf{X}_i]$$

Thus, any factor that affects the mean will also affect the variance.

This fact could be a potential drawback for using a Poisson regression model.

Poisson regression can also be used to model *rates* of events but we won't cover that case in this course.

Dataset

In this lecture, we'll use the [Bikeshare dataset](#) from the [ISL](#) package discussed in the [ISL book](#)

The dataset contains information on the hourly users of a bike sharing program in Washington DC, for different months, hours, weather conditions, and membership types (see [? Bikeshare](#))

The Bikeshare dataset has information about 8645 users

season	mnth	day	hr	holiday	weekday	workingday	weathersit	temp	atemp
3	Aug	220	6	0	1	1	clear	0.70	0.6
4	Oct	293	21	0	4	1	clear	0.44	0.4

A range problem

Since **counts are nonnegative integers**, if we use linear regression, we might have a range problem by predicting negative counts.

Instead, we can use a *function* of the linear component with a positive range, for example, an exponential function.

this is not the only function but it's commonly used due to its simplicity and interpretation

Poisson Regression

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The Poisson regression is defined as:

$$E[Y_i | \mathbf{X}_i] = \lambda_i = e^{\beta_0 + \beta_1 X_{1,i} + \dots + \beta_p X_{1,p}} \quad [\text{mean}]$$

or equivalently,

$$\log(\lambda_i) = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_p X_{1,p} \quad [\text{log-mean}]$$

the log function links the conditional expectation with the linear component in a natural way in the GLM family so it's called the *canonical link*.

We model the **log of the mean counts** with a linear component!!

then all that we learned for MLR can be used for this function of the conditional expectation!!

Note that there is no error term! we are modeling a function of the conditional expectation

Poisson: a generalized linear model

The Poisson regression also belongs to the generalized linear models (GLMs) family.

In R, we can use the `glm()` function again but setting `family = poisson` to calculate the *maximum likelihood estimator* of the coefficients.

`glm()` models, by default, the **log of the mean counts**, which is the “canonical link” function but other functions can be used

Fitting a Poisson Regression

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The interpretation of the coefficients will be very similar to those of the logistic regression case.

So this time let's fit a model with various types of covariates at once.

From all the covariates available, let's consider

- **workingday**: an indicator variable that equals 1 if it is neither a weekend nor a holiday
- **temp**: the normalized temperature, in Celsius
- **windspeed**: the normalized windspeed
- **season**: a qualitative variable that takes four values for each season

Factors vs numeric

Note that R will create dummy variables of factors, not of numerical variables.

sometimes numerical values are used for qualitative variables, e.g., season.

```
# A tibble: 2 × 5
  season workingday  temp windspeed bikers
  <dbl>      <dbl> <dbl>    <dbl>   <dbl>
1     1         0  0.26    0.134    34
2     1         1  0.28    0.104    96
```

You will need to adjust that before fitting the model

Let's fit a Poisson model to the data as is.

```
1 summary(glm(formula = bikers ~ ., data = bikeshare,
2             family = 'poisson'))
```

Call:

```
glm(formula = bikers ~ ., family = "poisson", data = bikeshare)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.411649	0.004158	820.39	<2e-16 ***
season	0.123221	0.000969	127.16	<2e-16 ***
workingday	-0.028971	0.001940	-14.93	<2e-16 ***
temp	2.050466	0.004919	416.82	<2e-16 ***
windspeed	0.822224	0.007361	111.70	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 1052921 on 8644 degrees of freedom
Residual deviance: 805356 on 8640 degrees of freedom
AIC: 858404

.. .. .

Note that **workingday** and **season** were fit as numerical variables, and both should be treated as categorical variables!!

R can't distinguish continuous from categorical variables if the variable is numeric.

Factors are required for qualitative variables!

```
1 bikeshare <- bikeshare %>%  
2   mutate(season = as.factor(season),  
3          workingday = as.factor(workingday))
```

Call:

```
glm(formula = bikers ~ ., family = "poisson", data = bikeshare)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.345358	0.003981	840.38	<2e-16	***
season2	0.062916	0.003741	16.82	<2e-16	***
season3	-0.141267	0.004320	-32.70	<2e-16	***
season4	0.379421	0.003357	113.01	<2e-16	***
workingday1	-0.036254	0.001942	-18.67	<2e-16	***
temp	2.688047	0.007435	361.54	<2e-16	***
windspeed	0.741449	0.007377	100.50	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

log-mean vs mean counts

As before, we can interpret the “raw” results corresponding to log-mean counts and are interpreted as for MLR

$$\log(E[\text{bikers}|\mathbf{X}]) = \beta_0 + \beta_1 S2 + \beta_2 S3 + \beta_3 S4 + \\ + \beta_4 W1 + \beta_5 \text{temp} + \beta_6 \text{wind}$$

for simplicity, S = season, W = workingday, and **wind** = windspeed, and $\mathbf{X}$ = all covariates

Interpretation: log-mean

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term	estimate	std.error	statistic	p.value
(Intercept)	3.345	0.004	840.384	0
season2	0.063	0.004	16.818	0
season3	-0.141	0.004	-32.699	0
season4	0.379	0.003	113.008	0
workingday1	-0.036	0.002	-18.672	0
temp	2.688	0.007	361.541	0
windspeed	0.741	0.007	100.505	0

iClicker: true or false

1. Keeping other variables in the model constant, at any value, the mean bike usage decrease 0.036 in working days
2. Keeping other variables in the model constant, at any value, a unit increase in the normalized temperature is associated with an increase in log mean bike usage of 2.688.

Interpretation: log-mean

The coefficients can be interpreted in terms of the log-means:

- **dummy variables** $\hat{\beta}_1 = 0.063$: the log-mean usage is season 2 is 0.063 larger than that in season 1 (reference), keeping all other variables in the model constant at any value
- **continuous variables** $\hat{\beta}_5 = 2.688$: an increase of 1 unit in **temp** is associated with an increase in the log-mean usage of 2.688, keeping all other variables in the model constant at any value

same as for MLR but with response “log-mean counts”!

Keeping constant ...

$$\log \lambda_{\text{temp}} = \beta_0 + \beta_5 \overbrace{\widehat{T}^{\text{temp}}}^{\text{Constant}} + \underbrace{\beta_1 S2 + \beta_2 S3 + \beta_3 S4 + \beta_4 W1 + \beta_6 \text{wind}}_{\text{Constant}}$$

$$\log \lambda_{\text{temp} + 1} = \beta_0 + \beta_5 \overbrace{(\widehat{T} + 1)^{\text{temp}+1}}^{\text{Constant}} + \underbrace{\beta_1 S2 + \beta_2 S3 + \beta_3 S4 + \beta_4 W1 + \beta_6 \text{wind}}_{\text{Constant}}$$

We take the difference between both equations as:

$$\begin{aligned} \log \lambda_{\text{temp} + 1} - \log \lambda_{\text{temp}} &= \beta_5 (\widehat{T} + 1) - \beta_5 \widehat{T} \\ &= \beta_5 \end{aligned}$$

Thus, we interpret these coefficients as **difference in log-mean counts**

Interpretation: mean counts

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But by logarithm property for a ratio:

$$\begin{aligned}\log \frac{\lambda_{\text{temp}} + 1}{\lambda_{\text{temp}}} &= \log \lambda_{\text{temp}} + 1 - \log \lambda_{\text{temp}} \\ &= \beta_5\end{aligned}$$

Finally, we have to exponentiate the previous equation:

$$\frac{\lambda_{\text{temp}} + 1}{\lambda_{\text{temp}}} = e^{\beta_5}$$

This expression indicates that an increase of 1 unit in a continuous covariate is associated with a mean count change **by a factor of e^{β_5}** (multiplicative change).

$$\lambda_{\text{temp}} + 1 = e^{\beta_5} \lambda_{\text{temp}}$$

A more natural way of interpreting the coefficients of a Poisson regression

Using `tidy`: `exponentiate = TRUE`

```
1 #Coefficients for the mean counts
2 tidy(bikers_poisson, exponentiate = TRUE) %>%
3   mutate_if(is.numeric, round, 3) %>%
4   knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	28.371	0.004	840.384	0
season2	1.065	0.004	16.818	0
season3	0.868	0.004	-32.699	0
season4	1.461	0.003	113.008	0
workingday1	0.964	0.002	-18.672	0
temp	14.703	0.007	361.541	0
windspeed	2.099	0.007	100.505	0

- **dummy variables** $e^{\hat{\beta}_1} = 1.065$: the mean usage is season 2 increases by a factor of 1.065 that in season 1 (reference), keeping all other variables in the model constant at any value
 - or by $(1.065 - 1) \times 100\% = 6.5\%$
- **continuous variables** $e^{\hat{\beta}_5} = 14.703$: an increase of 1 unit in **temp** is associated with an increase in mean usage of a factor of 14.703, keeping all other variables in the model constant at any value

iClicker:

dummy variables S3 $e^{\hat{\beta}_2} = -0.141 = 0.868$:

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A: a change in season from season 1 to season 3 is associated with a change in mean bike usage by a factor of 0.868, keeping all other variables in the model constant at any value

B: only 86.8% as many people will use bikes in season 3 relative to season season 1, keeping all other variables in the model constant at any value

C: a change in season from season 1 to season 3 is associated with a decrease of 13.2% in mean bike usage, keeping all other variables in the model constant at any value

—> **D:** all of the above are correct

Again, 4 Poisson regressions in 1!! For the **additive** model:

$$\log \lambda = \beta_0 + \beta_1 S2 + \beta_2 S3 + \beta_3 S4 + \beta_4 W1 + \beta_5 \text{temp} + \beta_6 \text{wind}$$

$$\log \lambda = \begin{cases} \beta_0 + \beta_4 W1 + \beta_5 \text{temp} + \beta_6 \text{wind} & \text{if } S2 = S3 = S4 = 0 \\ (\beta_0 + \beta_1) + \beta_4 W1 + \beta_5 \text{temp} + \beta_6 \text{wind} & \text{if } S2 = 1, S3 = S4 = 0 \\ (\beta_0 + \beta_2) + \beta_4 W1 + \beta_5 \text{temp} + \beta_6 \text{wind} & \text{if } S3 = 1, S2 = S4 = 0 \\ (\beta_0 + \beta_3) + \beta_4 W1 + \beta_5 \text{temp} + \beta_6 \text{wind} & \text{if } S4 = 1, S2 = S3 = 0 \end{cases}$$

(optional but useful) try to write these in exponential form

Model with interaction

For simplicity, let's look at a model with only **workingday** and **temp** but interacting

we are thinking that the “effect” of temperature will be different for leisure days compared to working days.

$$\log \lambda = \gamma_0 + \gamma_1 W1 + \gamma_2 \text{temp} + \gamma_3 W1 \times \text{temp}$$

Again, 2 Poisson regressions in 1 with **different** slopes!

$$\log \lambda = \gamma_0 + \gamma_1 W1 + \gamma_2 \text{temp} + \gamma_3 W1 \times \text{temp}$$

For the **log-mean counts**:

$$\log \lambda = \begin{cases} \gamma_0 + \gamma_2 \text{temp} & \text{if } W1 = 0 \\ (\gamma_0 + \gamma_1) + (\gamma_2 + \gamma_3) \text{temp} & \text{if } W1 = 1 \end{cases}$$

different *slope* (γ_2 vs $(\gamma_2 + \gamma_3)$), different *intercepts* (γ_0 vs $(\gamma_0 + \gamma_1)$)!

For the **mean counts**:

$$\lambda = \begin{cases} e^{\gamma_0} e^{\gamma_2 \text{temp}} & \text{if } W1 = 0 \\ e^{(\gamma_0 + \gamma_1)} e^{(\gamma_2 + \gamma_3) \text{temp}} & \text{if } W1 = 1 \end{cases}$$

in R

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```
1 bikers_poisson_int <- glm(bikers~workingday*temp,  
2                           data = bikeshare,  
3                           family = poisson)  
4  
5 tidy(bikers_poisson_int, exponentiate = TRUE) %>%  
6   mutate_if(is.numeric, round, 3) %>%  
7   knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	34.343	0.005	687.857	0
workingday1	1.481	0.006	63.450	0
temp	14.723	0.008	318.215	0
workingday1:temp	0.483	0.010	-71.201	0

Interpretation (for mean counts)

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In a **model with interactions**, the expected change in one variable **depends on** the value or levels the other variable

We **CAN'T** say “keeping the **temp** value constant” or “for working days or leisure days”

- *for leisure days*, a 1 unit increase in temperature is associated with an increase of mean bike usage **by a factor of** 14.723.

Interaction term

To get the “effect” of temperature for working days (i.e., `workingday` = 1), you need to *multiply* (instead of summing) the estimated coefficient of `temp` by that of the interaction term:

From the `tidy` table, with `exponentiate = TRUE`.

$$e^{\hat{\gamma}_2} = 14.723, e^{\hat{\gamma}_3} = 0.483$$

We need to multiply (instead of summing):

$$e^{\hat{\gamma}_2 + \hat{\gamma}_3} = e^{\hat{\beta}_2} e^{\hat{\beta}_3} = 14.723 \times 0.483 = 7.11$$

In working days, a unit increase in temperature is associated with an increase of mean usage **by a factor of** 7.11.

Inference

Inference for Poisson regression is the same as that of Logistic regression. They are based on theoretical results from the CLT.

if the sample size is large, the *Normal* distribution is used to obtain boundaries of confidence intervals and p -values

$$\frac{\hat{\beta}_j}{\hat{SE}(\hat{\beta}_j)} \sim N(0, 1)$$

This is called the **Wald's statistic**

The confidence intervals can be calculated for both the *log-mean* and the *mean* counts.

But again, R will do all this for you!

```
1 tidy(bikers_poisson,
2     conf.int = TRUE, conf.level = 0.9) %>%
3   mutate_if(is.numeric, round, 3) %>%
4   knitr::kable()
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	3.345	0.004	840.384	0	3.339	3.352
season2	0.063	0.004	16.818	0	0.057	0.069
season3	-0.141	0.004	-32.699	0	-0.148	-0.134
season4	0.379	0.003	113.008	0	0.374	0.385
workingday1	-0.036	0.002	-18.672	0	-0.039	-0.033
temp	2.688	0.007	361.541	0	2.676	2.700
windspeed	0.741	0.007	100.505	0	0.729	0.754

Residuals

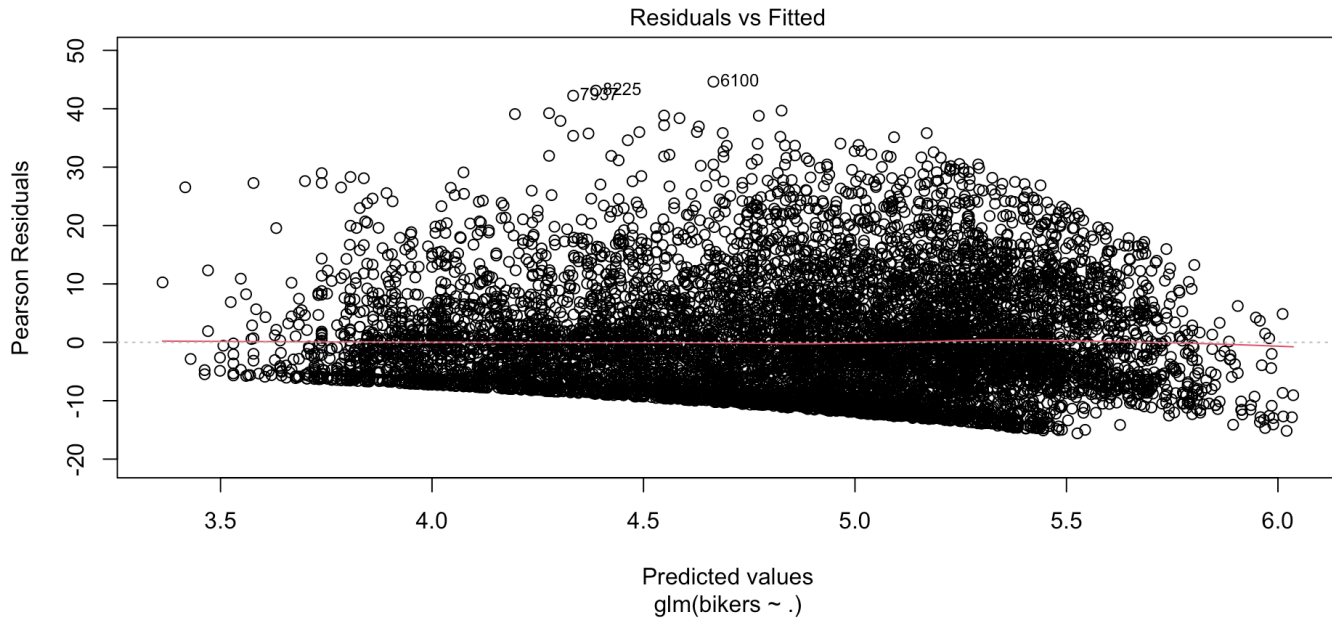
The Poisson regression has the same problems as the Logistic Regression:

- responses are discrete (at least more than 2 values now)
- the variance is not constant

$$\text{resid}_{\text{raw}} = y_i - \hat{\lambda}_i, \quad \text{resid}_{\text{pearson}} = \frac{y_i - \hat{\lambda}_i}{\sqrt{\hat{\lambda}_i}}$$

Residual plot

```
1 plot(bikers_poisson,1)
```



Overdispersion

The variance of a Poisson random variable equals the mean, which in practice may not be observed.

Poisson regression usually exhibits overdispersion.

An easy way to check is to fit a model using the quasiliikelihood method and check if the dispersion parameter is very different from 1.

in R

```
1 bikers_quasipoisson <- glm(bikers ~ .,  
2                             data = bikeshare,  
3                             family = quasipoisson)  
4  
5 summary(bikers_quasipoisson)
```

Call:

```
glm(formula = bikers ~ ., family = quasipoisson, data = bikeshare)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.34536	0.03789	88.286	< 2e-16	***
season2	0.06292	0.03561	1.767	0.077302	.
season3	-0.14127	0.04112	-3.435	0.000595	***
season4	0.37942	0.03196	11.872	< 2e-16	***
workingday1	-0.03625	0.01848	-1.962	0.049849	*
temp	2.68805	0.07077	37.981	< 2e-16	***
windspeed	0.74145	0.07022	10.558	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 90.60984)

Null deviance: 1052921 on 8644 degrees of freedom
Residual deviance: 783300 on 8638 degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5

Summary

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- A count response can be modelled using a Poisson Regression
- The (conditional) expectation of a count response is the (conditional) *mean counts*.
- The (conditional) expectation of a count response equals the (conditional) variance
- A LR can not be used to model the conditional expectation of a count response since its range extends beyond positive values
- Instead, one can model a function of the conditional mean counts. A common choice in Poisson regression is to use the *log* function

Summary (cont.)

The interpretation of the coefficients depends on the type of variables and the form of the model:

The raw coefficients (of the log-mean count model) are interpreted as:

- log-mean counts of a reference group
- difference of log-mean counts of a treatment vs a control group
- association between unit change in the input and changes in log-mean counts

Summary (cont.)

The **exponentiated coefficients** (of the mean count model) are interpreted as:

- mean counts of a reference group
- ratio of mean counts of a treatment vs a control group
- multiplicative changes in mean counts per unit change in the input

Summary (cont.)

- The estimated Poisson model can be used to make inference using the Wald's statistic: confidence intervals and tests
- There are different type of fitted values (log-mean counts or counts?) and residuals
- Overdispersion is a potential problem in Poisson regression. Using *quasiliikelihood* approach usual helps.